

# Deep Learning for Regional Downscaling of Summer Daily Maximum Temperature in the Northeastern United States

Xi Chen  
Brown University  
xi\_chen5@brown.edu

**Abstract**—High-resolution near-surface temperature fields are important for studying local heat exposure, but physically resolving climate at kilometer-scale resolution is computationally expensive. This project studies a supervised deep-learning approach for downscaling summer daily maximum 2-meter temperature over the northeastern United States. In simpler terms, the model starts with a coarse weather map where each grid cell is about 25 km wide and tries to produce a much more detailed 4 km temperature map. Rather than predicting temperature directly, the models learn the residual between GRIDMET maximum temperature and interpolated ERA5 maximum temperature. I compare linear interpolation, a convolutional U-Net, and a U-Net with a Transformer bottleneck. I also test whether adding dew point, a near-surface moisture variable often used as a measure of humidity, and static elevation improves performance. The deep-learning models substantially outperform interpolation: interpolation obtains 1.57°C RMSE, while the best Transformer model using temperature, dew point, and elevation obtains 1.14°C RMSE, a 27.6% improvement. The Transformer provides a modest gain over the U-Net, and dew point is most helpful on hot and humid days. Remaining errors are largest under cloudy and rainy conditions, suggesting that future models should include radiation, cloud, precipitation, land-cover, and vegetation information.

GitHub Repository: [https://github.com/xchen221/CSCI1470\\_DeepDownscale](https://github.com/xchen221/CSCI1470_DeepDownscale)

## I. Introduction

High-resolution summer daily maximum temperature is important because heat risk is highly local. Two places that are close together can experience different maximum temperatures because of elevation, distance from the coast, urban surfaces, and surrounding land cover. These local differences matter for heat-health analysis, outdoor work, agriculture, energy demand, and ecosystem stress; prior heat-health studies have used spatially resolved temperature data to estimate where harmful heat exposure occurs [1]. However, obtaining fine-resolution temperature fields is difficult. Running physical weather or climate models at kilometer-scale resolution over many years requires substantial computation, so high-resolution dynamical downscaling is often too expensive for large experiment sets [2]. This project asks whether a supervised deep-learning model can learn a cheaper coarse-to-fine mapping for summer maximum temperature.

The central problem in this project is regional statistical downscaling: given a coarse atmospheric estimate, predict a more spatially detailed daily maximum temperature field. I focus on the northeastern United States, including Rhode Island and surrounding states, during June, July, and August. The coarse predictor is ERA5 daily maximum 2-meter temperature at approximately 0.25° resolution, while the high-resolution target is GRIDMET-like 4 km maximum temperature. A simple interpolation baseline can place ERA5 on the target grid, but it cannot fully recover local gradients caused by topography, coastline geometry, or land-surface differences.

My approach combines residual learning with two neural architectures: a convolutional U-Net and a U-Net with a Transformer bottleneck. The main question is whether these models can learn fine-scale corrections beyond interpolation, and whether extra predictors such as elevation and dew point improve the correction.

The project goals evolved in three levels. The base goal was to build a clean preprocessing, training, and evaluation pipeline and beat linear interpolation with a convolutional model. The target goal was to compare a U-Net and a Transformer-based model on the same downscaling task. The stretch goal was to add physically meaningful predictors and diagnose when different model choices help. By the end, I reached the base and target goals and made progress on the stretch goal by adding dew point and elevation, then analyzing performance by weather regime.

## II. Literature Review

Statistical downscaling is a standard approach in climate-impact work because many impact questions require higher spatial resolution than global or regional climate products can provide directly [3]. Traditional downscaling and bias-correction methods are useful, but they often require strong assumptions about the relationship between coarse predictors and local outcomes. Deep learning is attractive for this problem because it can learn nonlinear, spatially structured corrections from data.

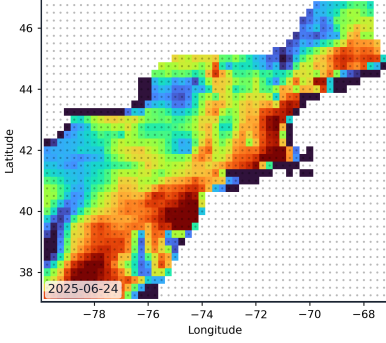
The U-Net architecture was originally developed for biomedical image segmentation [4], but its structure is well matched to gridded geophysical downscaling. The

## Data and Downscaling Task

Summer daily maximum 2 m temperature over the northeastern United States

### Input

ERA5 Tmax, 0.25 deg (~25 km)

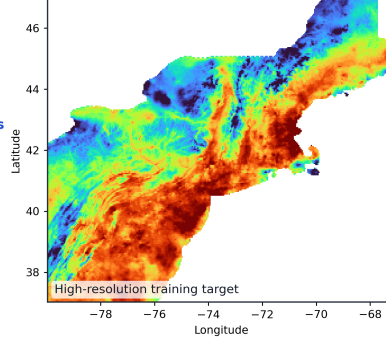


model learns

coarse-to-fine mapping

### Target

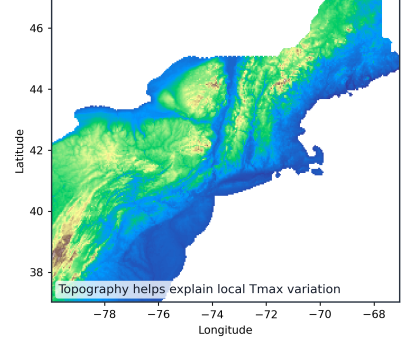
GRIDMET Tmax, 4 km



High-resolution training target

### Static Predictor

Elevation on 4 km grid



Topography helps explain local Tmax variation

Training split: 2000-2018 | validation: 2019-2021 | testing: 2022-2025

Fig. 1. Overview of the downscaling task. The model uses coarse ERA5 temperature information and static terrain information to learn a coarse-to-fine mapping toward a 4 km GRIDMET target field.

encoder compresses spatial information and expands the receptive field, the decoder reconstructs an output field at the original resolution, and skip connections preserve local detail. In this project, the U-Net plays the role of an image-to-image residual mapper: it receives predictor fields and returns one residual value per target grid cell.

Transformers introduced self-attention as a way to model long-range dependencies [5]. For climate and weather fields, this is appealing because temperature errors are not purely local. A coastal gradient, regional heat dome, or humid air mass can influence broad parts of the domain. However, full-resolution attention over every 4 km grid cell would be expensive. I therefore used a hybrid architecture: a U-Net encoder reduces the spatial grid, a Transformer operates only at the bottleneck, and a U-Net decoder reconstructs the residual field. This keeps the local strengths of convolutions while allowing broader spatial interaction.

The data sources also shaped the method. ERA5 is a widely used global atmospheric reanalysis [6]; GRIDMET provides high-resolution gridded surface meteorological fields over the contiguous United States [7]; and ETOPO supplies static terrain information [8]. These sources make it possible to train a supervised model that combines dynamic atmospheric variables with static geographic features.

### III. Methodology

#### A. Dataset and Preprocessing

The modeling domain covers the northeastern United States, roughly 37–47°N and 80–60°W. I used summer days only, since the scientific motivation is local heat exposure and because limiting the season reduces the amount of seasonal variability the model must learn. The

time period is 2000–2025. To avoid temporal leakage, I used a chronological split: 2000–2018 for training, 2019–2021 for validation, and 2022–2025 for testing.

The raw datasets did not start on the same grid or even the same valid spatial footprint. ERA5 temperature and ERA5 dew point are coarse 25 km fields, GRIDMET provides the target daily maximum temperature at about 4 km resolution, the urban fraction source is a 1 km raster, and ETOPO2 provides high-resolution static elevation/topographic information. Because the high-resolution temperature target is only available over the continental United States, I created a valid spatial mask from the target grid and applied it to all predictors and targets, excluding ocean cells, Canada, and any locations without target data. I then used linear interpolation to place each predictor onto the common 4 km GRIDMET latitude-longitude grid so every input channel had shape  $H \times W$  and described the same locations.

The baseline dynamic predictor is ERA5 daily maximum 2-meter temperature after interpolation to the GRIDMET grid. Rather than predicting the full high-resolution temperature field directly, the model predicts a residual correction:

$$r = T_{\text{GRIDMET}} - T_{\text{ERA5,interp}}. \quad (1)$$

The final downscaled field is reconstructed by adding the predicted residual back to the interpolated ERA5 field:

$$\hat{T}_{\text{highres}} = T_{\text{ERA5,interp}} + \hat{r}. \quad (2)$$

This residual setup makes the learning task more focused: the interpolated ERA5 field already contains the large-scale weather pattern, so the neural network can concentrate on the missing fine-scale correction. Additional predictors were added in later experiments. ERA5 dew

point is a dynamic humidity-related feature from the same ERA5 data source and the same coarse 25 km grid as ERA5 temperature. In contrast, elevation from ETOPO2 and urban fraction from a Zenodo land-surface dataset are static, higher-resolution surface features. All input features were first interpolated or aligned to the 4 km target grid using linear interpolation, and invalid cells were handled with the spatial mask.

## B. Models

I compared three model families. The first is not a learned model: linear interpolation from ERA5 to the 4 km grid. This is the reference point that any deep-learning model should beat.

The second model is a residual U-Net. Its input tensor has shape  $[B, C, H, W]$ , where  $B$  is batch size,  $C$  is the number of input channels, and  $H, W$  are the spatial dimensions. The model uses two 3-by-3 convolution layers with ReLU in each block. The encoder maps channels  $C \rightarrow 32 \rightarrow 64 \rightarrow 128$  while downsampling twice with max pooling. The decoder upsamples using transposed convolutions, concatenates skip connections from the encoder, and returns one residual channel through a final 1-by-1 convolution. The skip connections are important because downscaling needs both broad context and grid-cell-level detail.

The third model is a U-Net-Transformer. It begins like the U-Net, but adds one deeper downsampling stage so the bottleneck has lower spatial resolution. A 1-by-1 convolution projects the bottleneck features to a 256-dimensional embedding. I then add learnable 2D positional embeddings and apply a Transformer encoder with 3 layers, 8 attention heads, GELU activations, and 0.1 dropout. The output tokens are reshaped back to a feature map and passed through the decoder. This design applies self-attention where it is computationally manageable, while still using convolutions for local feature extraction and reconstruction.

## C. Training and Evaluation

All model inputs and residual targets were normalized using training-set statistics. During training and evaluation, the loss and metrics were computed only over valid cells, which prevented missing or out-of-domain locations from affecting gradients or reported performance.

The models were trained with Adam and masked mean squared error:

$$\mathcal{L} = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} (\hat{r}_{i,j} - r_{i,j})^2, \quad (3)$$

where  $\Omega$  is the set of valid grid cells. The plain U-Net used a batch size of 32, learning rate  $3 \times 10^{-4}$ , and 20 epochs. The Transformer models were heavier, so they used a batch size of 4, learning rate  $1 \times 10^{-4}$ , and 30 epochs. A ReduceLROnPlateau scheduler lowered the learning

TABLE I

Summary of major experiments. T denotes interpolated ERA5 daily maximum temperature, D2M denotes ERA5 dew point, Elev denotes elevation, and Urban denotes urban fraction.

| Experiment    | Inputs       | RMSE  | Corr. | Improv. |
|---------------|--------------|-------|-------|---------|
| Interpolation | T            | 1.569 | 0.941 | 0.0%    |
| U-Net         | T only       | 1.317 | 0.953 | 16.0%   |
| U-Net         | T+Elev       | 1.170 | 0.963 | 25.4%   |
| U-Net         | T+D2M+Elev   | 1.221 | 0.963 | 22.2%   |
| U-Net         | T+Elev+Urban | 1.196 | 0.964 | 23.8%   |
| Transformer   | T only       | 1.255 | 0.959 | 20.0%   |
| Transformer   | T+Elev       | 1.155 | 0.965 | 26.4%   |
| Transformer   | T+D2M+Elev   | 1.136 | 0.966 | 27.6%   |
| Transformer   | T+Elev+Urban | 1.158 | 0.965 | 26.2%   |

rate when validation loss stopped improving, and the best validation checkpoint was saved for testing.

I evaluated performance with RMSE, bias, and spatial correlation. RMSE measures error magnitude, bias measures systematic warm or cold error, and correlation measures whether the predicted spatial pattern matches the target field. The primary metric is RMSE because large temperature errors are especially important for heat-risk applications.

## IV. Results

### A. Overall Performance

The deep-learning models clearly improved over interpolation. Table I summarizes the main experiments, including the simpler ablations and the final richer models. Interpolation had an RMSE of  $1.57^\circ\text{C}$  and a spatial correlation of 0.941. Even the temperature-only U-Net improved over interpolation, and adding elevation produced a much stronger U-Net baseline. The best model was the Transformer with temperature, dew point, and elevation, which reached  $1.14^\circ\text{C}$  RMSE and spatial correlation 0.966.

The comparison also shows that architecture was not the only factor. Elevation helped both model families, while dew point behaved differently across architectures. Adding dew point made the U-Net worse than the simpler temperature-plus-elevation U-Net, but it improved the Transformer model. One interpretation is that dew point is a coarse, larger-scale weather feature, unlike elevation and urban fraction, which provide more local surface information. The plain U-Net may not use that broad context effectively, while the Transformer bottleneck can integrate it with regional spatial information. Figure 3 shows the spatial RMSE patterns behind these aggregate scores.

The urban-fraction experiments were motivated by the urban heat-island effect, but they did not improve performance as much as expected. There are several possible reasons. First, the current U-Net and bottleneck-Transformer models may be too simple to learn the full interaction between urban surfaces, surrounding land cover, and atmospheric conditions. Second, urban fraction may be an incomplete proxy for urban heat: two

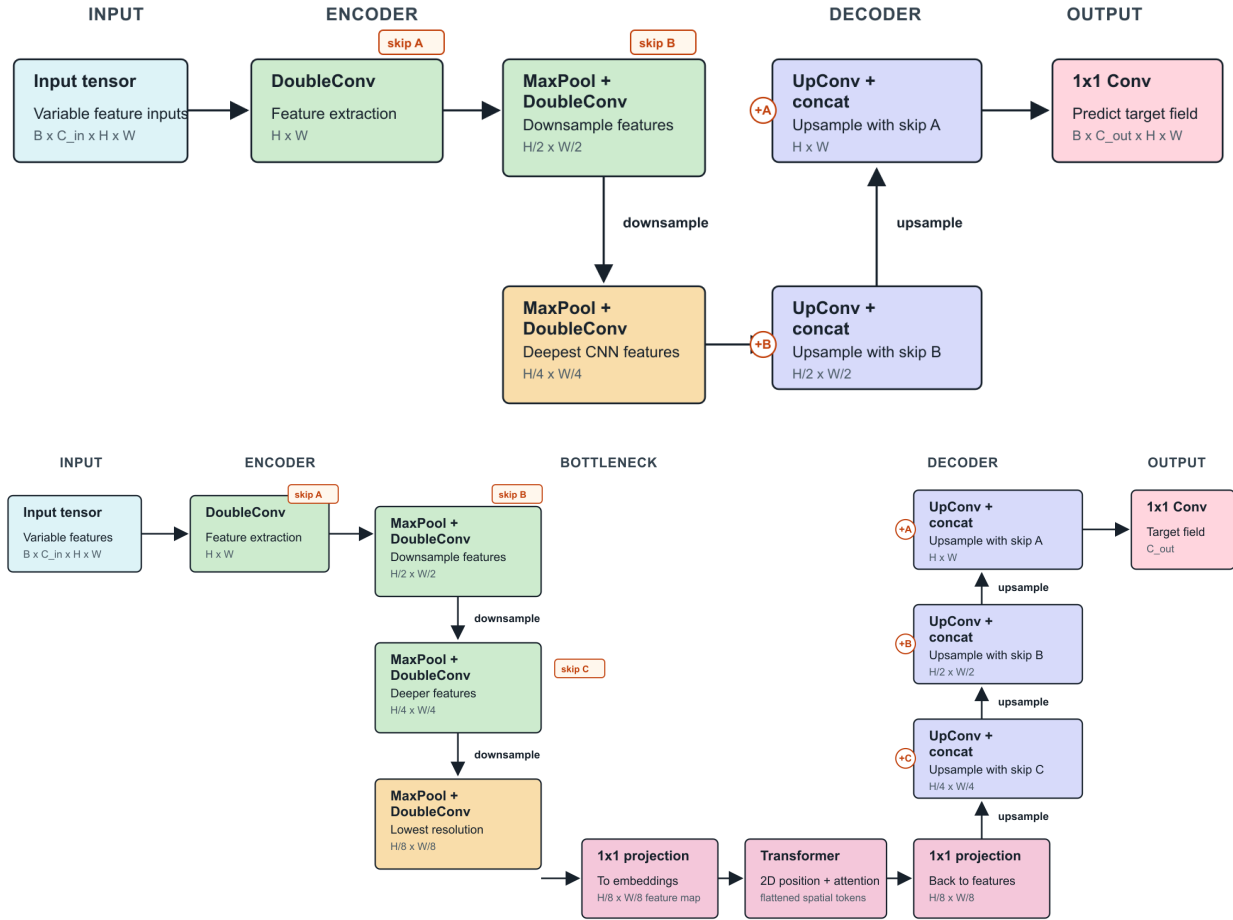


Fig. 2. Model architecture schematics. The U-Net uses convolutional encoding, decoding, and skip connections to learn local residual corrections. The U-Net-Transformer keeps the same encoder-decoder idea but inserts a Transformer bottleneck at low spatial resolution, allowing attention over broader regional context without applying attention to every full-resolution grid cell.

grid cells with similar urban fraction can have different building density, vegetation, impervious surface, or land-use type. A categorical land-cover input, vegetation index, or impervious-surface feature might describe urban heat effects more directly. Finally, urban effects may require models with stronger spatial-context modeling, such as a larger Transformer or a full attention-based architecture, although that would increase data and compute requirements.

## B. Effect of Dew Point

Dew point did not help uniformly across all days. When comparing the Transformer with temperature and elevation against the Transformer with temperature, dew point, and elevation, the dew-point model improved most on hot and humid regimes (Figure 4). On hot days, dew point reduced RMSE by about  $0.13^{\circ}\text{C}$ . On humid days, it reduced RMSE by about  $0.09^{\circ}\text{C}$ . On cool, dry, and moderate days, the dew-point feature helped less and sometimes slightly hurt performance.

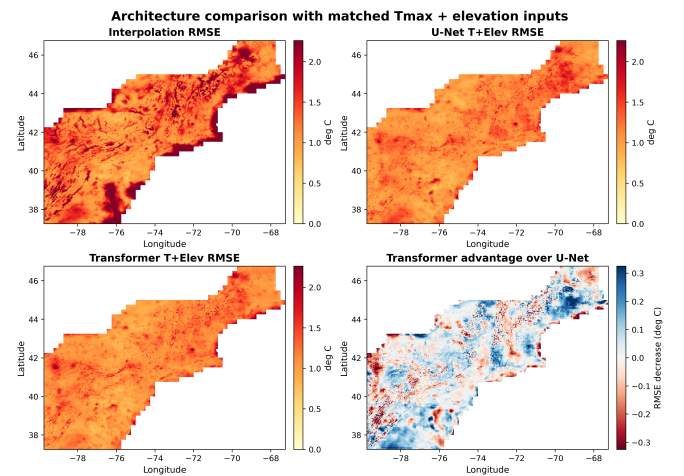


Fig. 3. Spatial RMSE comparison across interpolation, U-Net, and Transformer-style downscaling models. Deep learning reduces error relative to interpolation, with the best Transformer model giving the lowest overall RMSE.

This result is physically sensible. Dew point carries information about air mass and humidity structure, which is especially relevant during hot and humid summer conditions. On days when humidity is not tightly connected to the residual temperature pattern, the extra channel can act more like weak or noisy context. The result also suggests that adding predictors should be done thoughtfully: more variables are not automatically better unless they explain the specific errors the model is making.

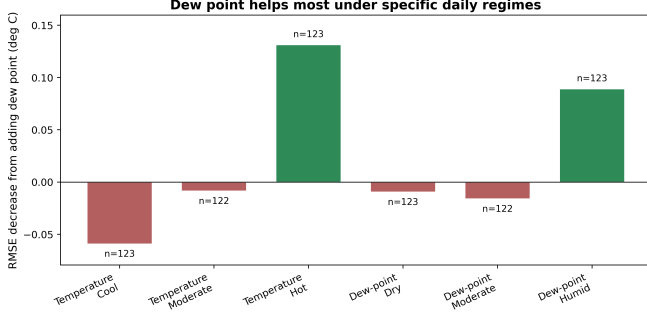


Fig. 4. Dew point provides the clearest benefit under hot and humid conditions, suggesting that additional meteorological predictors can help when they are physically tied to the residual structure.

The spatial pattern in Figure 5 also suggests why dew point may be useful in some parts of the domain. Benefits appear in several regions where humidity and local temperature are likely coupled, including parts of western New York and the Allegheny Plateau, coastal corridors, and some terrain-influenced areas. This may reflect larger-scale moisture gradients associated with nearby water bodies such as the Great Lakes and the Atlantic coast. I treat this as a hypothesis rather than a confirmed mechanism, because the current experiments did not explicitly isolate lake influence, coastline distance, or wind direction.

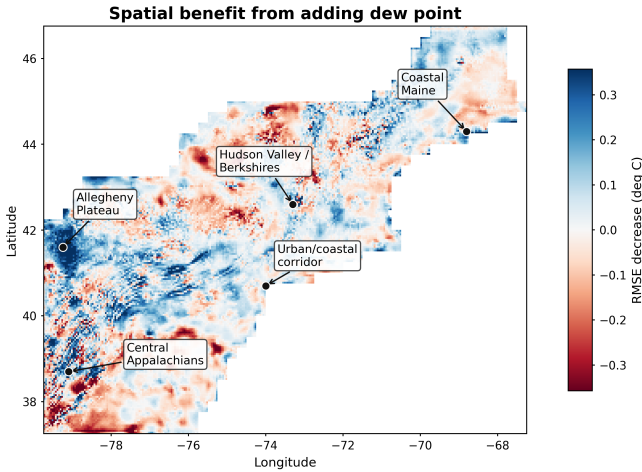


Fig. 5. Spatial pattern of the Transformer RMSE change after adding dew point. Blue indicates locations where dew point reduced RMSE, while red indicates locations where it increased RMSE.

### C. Failure Modes

The best model still struggled under cloudy and rainy conditions (Figure 6). In the weather-regime summary, very high cloud cover produced mean RMSE around  $1.25^{\circ}\text{C}$ , high cloud cover around  $1.22^{\circ}\text{C}$ , and high precipitation around  $1.19^{\circ}\text{C}$ , all worse than the all-day average near  $1.09^{\circ}\text{C}$  in that diagnostic table. In contrast, hot days and hot-humid days were not the worst cases for the final model.

This pattern points to a missing-information problem rather than only an architecture problem. Clouds and precipitation affect surface heating through radiation, soil moisture, and boundary-layer processes. My final inputs include temperature, dew point, and elevation, but not cloud cover, shortwave radiation, precipitation, vegetation, or detailed land cover. The model therefore has limited information for explaining why two places with similar coarse temperature and terrain might have different local heating under cloudy or rainy conditions. Another possible factor is input quality: the model starts from the coarse 25 km ERA5 temperature field, so days with larger ERA5 biases or missing physical structure give the residual model a harder correction problem.

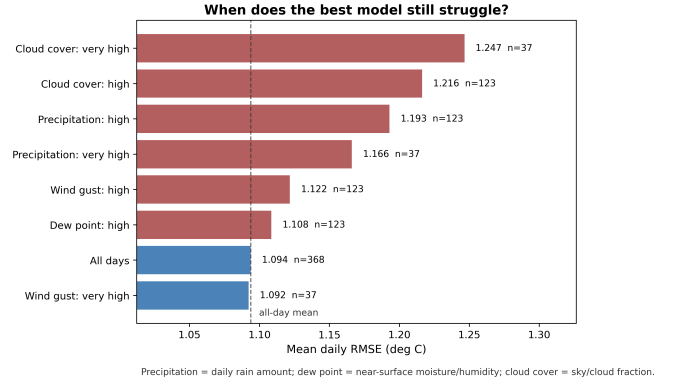


Fig. 6. Weather-regime diagnostics for the best model. Cloudy and rainy regimes have higher mean daily RMSE than the all-day average, suggesting that cloud, radiation, and precipitation predictors would be useful future additions.

### D. Discussion and Implications

The results suggest that residual deep learning is a useful way to improve coarse temperature fields, but they also show that the input variables matter as much as the architecture. Static surface features such as elevation provide local information that both U-Net and Transformer models can use. Coarser dynamic features such as dew point are more conditional: they help when humidity is tied to the error structure, especially for the Transformer, but they do not automatically improve every model. For practical downscaling, this means future work should prioritize physically relevant predictors rather than simply adding more channels. The failure-mode analysis also points to two important limits: missing predictors such as

cloud cover, shortwave radiation, precipitation, vegetation, land-cover categories, and more detailed urban-surface variables; and the quality of the original coarse ERA5 temperature input.

## V. Challenges

The largest challenge was data engineering. The project required aligning multiple datasets with different native resolutions, coordinate conventions, time coverage, and missing-value patterns. A small mismatch in latitude, longitude, or time would silently corrupt the supervised learning problem because the model would be asked to map one day or location to another. I spent a meaningful amount of the project building preprocessing notebooks and dataset classes to make sure ERA5, GRIDMET, dew point, elevation, and masks were consistently aligned.

A second challenge was scope control. My original plan included CNNs, Transformers, static features, and weather-regime analysis. That is a lot for one final project, and the pipeline had to work before any comparison was meaningful. I therefore moved in stages: first interpolation and a plain U-Net, then Transformer experiments, then extra predictors and diagnostics. This was the right order because it gave me a working baseline early, even though it meant some stretch ideas, such as richer land-cover variables, stayed as future work.

The third challenge was computation. The Transformer model was more expensive than the plain U-Net, which forced a smaller batch size and more careful training. I eventually needed to run the heavier Transformer experiments on Oscar, and a significant amount of time went into configuring the environment, matching package versions, and making the training scripts run reliably there. I also had to keep attention at the bottleneck rather than the full-resolution grid. This limitation ended up becoming a useful architectural choice: the hybrid model is not the most ambitious Transformer possible, but it is a practical compromise between spatial context and trainability.

## VI. Ethics

The positive motivation for this project is clear: better local temperature information can support heat-risk assessment, public health planning, infrastructure design, and environmental research. However, downscaled climate and weather products also carry risks. A model that looks accurate on average can still fail in particular places or conditions, and those failures may matter most for vulnerable communities. For example, underestimating maximum temperature during a cloudy but humid heat event could lead to false confidence if the output were used operationally.

There are also fairness issues. Temperature exposure is shaped by housing, tree cover, work conditions, air conditioning access, and urban design. My model uses gridded meteorological and surface variables, not social vulnerability data. That is appropriate for the technical

scope of this project, but it means the output should not be treated as a complete heat-risk product. A future applied version should evaluate errors across urban, coastal, rural, and disadvantaged communities, not only across aggregate RMSE.

Privacy concerns are low because the datasets are public gridded environmental products rather than individual-level data. Environmental impact is more relevant: deep learning experiments consume compute, and climate applications should be honest about that cost. I mitigated this by using relatively small models, chronological splits, and a limited number of final experiments rather than training very large architectures without a clear purpose. Finally, the model should be communicated as a research prototype. It improves interpolation, but it is not a replacement for physical understanding or operational forecast validation.

## VII. Reflection

Overall, I feel the project turned out well. The base goal was met because the learned models beat interpolation by a large margin. The target goal was met because I compared a plain U-Net and a U-Net-Transformer under a shared residual-learning setup. The stretch goal was partly met: I added dew point and elevation and analyzed when dew point helps, but I did not fully explore cloud, radiation, precipitation, vegetation, or land-cover predictors.

The model worked broadly the way I expected, but the size of the improvements was interesting. I expected deep learning to beat interpolation, and it did. I also expected the Transformer to dominate the U-Net more strongly than it did. Instead, the improvement from U-Net to Transformer was modest. My interpretation is that local convolutional structure is already very powerful for this downscaling task, and the bottleneck attention mostly provides incremental help for broader spatial organization.

My approach changed over time from "compare architectures" to "understand what information the model is missing." Early in the project, I thought the architecture comparison would be the main story. By the end, the more interesting story was conditional performance: dew point helps on hot and humid days, while cloudy and rainy days remain hard. That pivot made the results feel more scientific, because the analysis connected model errors back to physical weather conditions.

If I could do the project again, I would build the predictor set earlier and more systematically. I would include cloud cover, shortwave radiation, precipitation, vegetation indices, land-cover categories, and a more careful urban feature from the beginning. I would also reserve more time for ablation studies so each added variable could be tested cleanly. With more time, I would evaluate performance by geography, especially coastal areas, high-elevation regions, and urban grid cells, and I would test whether the model generalizes beyond the Northeast or beyond summer.

My biggest takeaway is that deep learning for environmental data is as much about the data pipeline and problem formulation as it is about the model. The residual-learning setup mattered. The mask mattered. The chronological split mattered. The model architecture mattered too, but it was only one part of the full system. I also learned that model diagnostics are more satisfying than a single leaderboard number: seeing when the model fails gives a much clearer path for future improvement.

### VIII. Conclusion

This project built a residual deep-learning pipeline for downscaling summer daily maximum 2-meter temperature over the northeastern United States. Compared with linear interpolation, both the U-Net and U-Net-Transformer learned meaningful fine-scale corrections. The best model, a U-Net-Transformer using temperature, dew point, and elevation, reduced RMSE from  $1.57^{\circ}\text{C}$  to  $1.14^{\circ}\text{C}$  and improved spatial correlation from 0.941 to 0.966. The main gain came from learned residual correction, while the Transformer and dew point added smaller but useful improvements.

The most important limitation is that the current predictor set is incomplete. Cloudy and rainy conditions remain difficult, likely because the model does not yet include cloud, radiation, precipitation, vegetation, or detailed land-surface variables. Future work should focus less on making the architecture larger and more on adding physically relevant predictors and evaluating errors across meaningful weather regimes and communities.

### Acknowledgments

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